

Quotient Laws of Exponents

MA.7.NSO.1.1

I can use the quotient of powers, power of a quotient and zero exponent laws to rewrite and evaluate a numerical expression.

Yesterday: same base, add the exponents.

Today: same base, subtract — and exponent 0 is not 0.

Warm-Up

On your own. Four minutes. No notes.

1. Write as one power: $3^2 \cdot 3^3$
2. $\sqrt[3]{-27}$
3. Solve $2x + 3 < 11$
4. Simplify $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$

Warm-Up

Answers.

1. Write as one power: $3^2 \cdot 3^3$ 3^5
2. $\sqrt[3]{-27}$ -3
3. Solve $2x + 3 < 11$ $x < 4$
4. Simplify $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$ 5

Notes

Where the rule comes from.

I. Quotient of powers — same base, subtract the exponents

$$\frac{5^6}{5^3} = \frac{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}{5 \cdot 5 \cdot 5} = 5 \cdot 5 \cdot 5 = 5^3$$

Quotient of powers: $\frac{a^m}{a^n} = a^{m-n} \quad (a \neq 0)$

Check with values: $\frac{5^6}{5^3} = \frac{15625}{125} = 125 = 5^3$

Notes

II. Power of a quotient — the exponent goes top and bottom

$$\left(\frac{8}{10}\right)^3 = \frac{8}{10} \cdot \frac{8}{10} \cdot \frac{8}{10} = \frac{8 \cdot 8 \cdot 8}{10 \cdot 10 \cdot 10} = \frac{8^3}{10^3}$$

Power of a quotient: $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \quad (b \neq 0)$

Either way: $\frac{8^3}{10^3} = \frac{512}{1000} = 0.512$ and $(0.8)^3 = 0.512$

Notes

III. The zero exponent — two roads to the same answer

2^4	2^3	2^2	2^1	2^0
16	8	4	2	1

$$\frac{2^3}{2^3} = \frac{8}{8} = 1 \quad \text{and} \quad \frac{2^3}{2^3} = 2^{3-3} = 2^0 \quad \text{so} \quad 2^0 = 1$$

Zero exponent: $a^0 = 1$ for every nonzero a

Example 1

A negative base.

$$\frac{(-2)^6}{(-2)^3}$$

Which law? Then find the value.

Example 1

Worked.

$$\frac{(-2)^6}{(-2)^3} = (-2)^{6-3} = (-2)^3 \quad \text{quotient of powers: subtract}$$

$$(-2)^3 = -8 \quad \text{three negative factors}$$

$$64 \div (-8) = -8 \quad \text{check: evaluate each, then divide}$$

Answer: -8

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{3^6}{3^2}$$

Your Turn

Answer.

$$\frac{3^6}{3^2}$$

same base, subtract: 3^4

Answer: $3^4 = 81$

Example 2

A decimal base inside a quotient.

$$\left(\frac{0.2}{10}\right)^4$$

Find the value.

Example 2

Worked both ways.

$$\left(\frac{0.2}{10}\right)^4 = \frac{0.2^4}{10^4} = \frac{0.0016}{10000} \quad \text{power of a quotient}$$

$$\frac{0.0016}{10000} = 0.00000016 \quad \text{divide by } 10^4: \text{four places left}$$

$$(0.02)^4 = 0.00000016 \quad \text{check: divide first, then raise}$$

Answer: 0.00000016

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(\frac{3}{4}\right)^3$$

Your Turn

Answer.

$$\left(\frac{3}{4}\right)^3$$

numerator and denominator both cubed

Answer: $\frac{27}{64}$

Whiteboards • Question 1 of 9

Boards up on three.

$$\frac{10^5}{10^3}$$

Answer it.

Write it as one power.

Whiteboards • Question 1 of 9

Answer.

$$\frac{10^5}{10^3}$$

same base, subtract the exponents

Answer: 10^2

Whiteboards • Question 2 of 9

Boards up on three.

A server stores 8^4 files in 8^1 folders, evenly.

It has been running for 30 days.

How many files are in each folder?

Answer it.

Find the value.

Whiteboards • Question 2 of 9

Answer.

A server stores 8^4 files in 8^1 folders, evenly.

It has been running for 30 days.

How many files are in each folder?

$$8^4 \div 8^1 = 8^3$$

Answer: 512

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{(-6)^6}{(-6)^5}$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$\frac{(-6)^6}{(-6)^5}$$
$$(-6)^1$$

Answer: -6

Whiteboards • Question 4 of 9

Boards up on three.

$$\left(\frac{-5}{10}\right)^7$$

Answer it.

Apply the power of a quotient law. Do not simplify.

Whiteboards • Question 4 of 9

Answer.

$$\left(\frac{-5}{10}\right)^7$$

the exponent goes to both

Answer: $\frac{(-5)^7}{10^7}$

Whiteboards • Question 5 of 9

Boards up on three.

$$9^0$$

Which is the value?

A. 1

B. 0

C. 9

D. $\frac{1}{9}$

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$9^0$$

Which is the value?

A. 1

B. 0

C. 9

D. 1/9

Answer: A — 1

Whiteboards • Question 6 of 9

Boards up on three.

$$\frac{4^8}{4^2}$$

Which expression is equivalent?

A. 4^6

B. 4^4

C. 1^6

D. 4^{10}

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\frac{4^8}{4^2}$$

Which expression is equivalent?

A. 4^6

B. 4^4

C. 1^6

D. 4^{10}

Answer: A — 4^6

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{0.5^8}{0.5^5}$$

Answer it.

Find the value.

Whiteboards • Question 7 of 9

Answer.

$$\frac{0.5^8}{0.5^5}$$
$$0.5^3$$

Answer: 0.125

Whiteboards • Question 8 of 9

Boards up on three.

$$\left(\frac{10}{11}\right)^0$$

Answer it.

Find the value.

Whiteboards • Question 8 of 9

Answer.

$$\left(\frac{10}{11}\right)^0$$

any nonzero base to the zero power

Answer: 1

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Hannah cancels: $\frac{2^3}{2^3} = \frac{8}{8} = 1$. Sebastian subtracts: $\frac{2^3}{2^3} = 2^0$.

Who is right?

Write your answer in sentences.

Say what each one did, and what it tells you about 2^0 .

Whiteboards • Question 9 of 9

Answer.

Hannah cancels: $\frac{2^3}{2^3} = \frac{8}{8} = 1$. Sebastian subtracts: $\frac{2^3}{2^3} = 2^0$.

Who is right?

both are right, so 2^0 must equal 1

Answer: Both — $2^0 = 1$

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Powers with negative bases — WFW

Optional: Division with exponents — XNC

Optional: Identify equivalent expressions involving whole-number exponents — KBE